№5

a) $\xi \sim g(x) = \frac{1}{\theta}\{(\theta;2\theta)\}$

$$M\xi = \int\limits_{\theta}^{2\theta} x\cdot\frac{1}{\theta}\,dx = \frac{1}{\theta}\cdot\left(\frac{4\theta^2}{2}-\frac{\theta^2}{2}\right) = \frac{3\theta}{2}$$

$$M\xi^2 = \int\limits_{\theta}^{2\theta} x^2\cdot\frac{1}{\theta}\,dx = \frac{1}{\theta}\left(\frac{8\theta^3}{3}-\frac{\theta^3}{3}\right) = \frac{7\theta^2}{3}$$

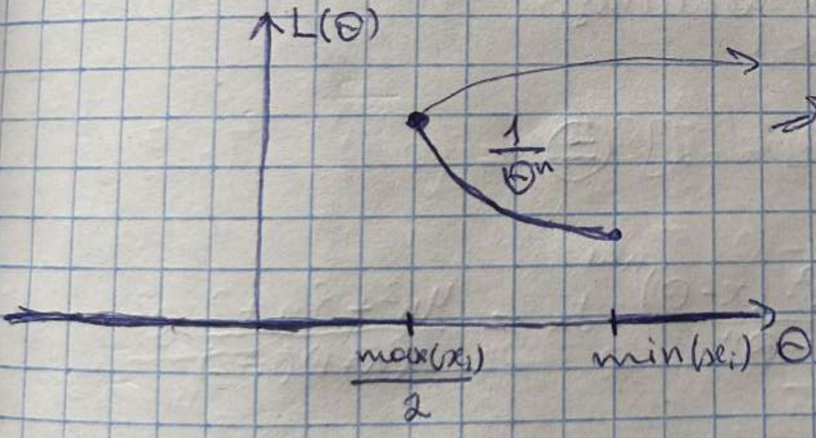
$$D\xi = M\xi^2-(M\xi)^2 = \frac{7\theta^2}{3}-\frac{9\theta^2}{4} = \frac{28\theta^2-27\theta^2}{12} = \frac{\theta^2}{12}$$

<u>ОММ</u>: $\alpha_1 = \tilde{\alpha}_1 = \frac{1}{n}\sum\limits_{i=1}^{n} x_i = \overline{x} \Rightarrow \frac{3\theta}{2} = \overline{x}$

$$\Rightarrow \tilde{\theta}_1 = \frac{2\overline{x}}{3}$$

ОМП: $L(\theta) = \prod\limits_{i=1}^{n} g(x_i,\theta) = \frac{1}{\theta^n}\{\forall i: \theta < x_i < 2\theta\} =$

$$= \frac{1}{\theta^n}\begin{cases}\theta > \frac{\max(x_i)}{2}\\ \theta < \min(x_i)\end{cases}$$



$$\Rightarrow \tilde{\theta}_2 = \frac{x_{max}}{2}$$

б) $\tilde{\Theta}_1$: $\tilde{\Theta}_1 = \frac{2}{3}\bar{x}$

$$M\tilde{\Theta}_1 = \frac{2}{3} M\bar{x} = \frac{2}{3}\cdot\frac{1}{n} M\left(\sum_{i=1}^{n} x_i\right) = \frac{2}{3n}\cdot n M\xi = \Theta \Rightarrow$$

$\Rightarrow$ несмещ.

$$D\tilde{\Theta}_1 = D\left[\frac{2}{3}\bar{x}\right] = \frac{4}{9}\cdot\frac{1}{n^2}\cdot n\cdot D\xi = \frac{1}{n}\cdot\frac{\Theta^2}{27} \xrightarrow[n\to\infty]{} 0 \Rightarrow$$

$\Rightarrow$ состоят.

$\tilde{\Theta}_2$: $\tilde{\Theta}_2 = \frac{x_{max}}{2}$; $x_{max} \sim \underbrace{(F(x))^n}_{\Psi(x)}$

$$M\tilde{\Theta}_2 = \frac{1}{2} M x_{max} = \left\{ \begin{array}{l} \Psi = \left(\frac{1}{\Theta}\int_{\Theta}^{x} dx\right)^n = \left(\frac{x-\Theta}{\Theta}\right)^n \\ \Psi'(x) = \frac{n(x-\Theta)^{n-1}}{\Theta^n}\{(\Theta, 2\Theta)\} \end{array} \right\} =$$

$$= \frac{1}{2}\int_{\Theta}^{2\Theta} \frac{x\cdot n(x-\Theta)^{n-1}}{\Theta^n}\,dx = \frac{1}{2\Theta^n}\int_{\Theta}^{2\Theta} n\cdot x(x-\Theta)^{n-1}\,dx =$$

$$= \frac{1}{2\Theta^n}\left(2\Theta^{n+1} - \frac{\Theta^{n+1}}{n+1}\right) = \Theta - \frac{\Theta}{2(n+1)} = \frac{\Theta(2n+1)}{2n+2} \Rightarrow \text{смещ.}$$

$\tilde{\Theta}_2' = \frac{2n+2}{2n+1}\tilde{\Theta}_2 \Rightarrow$ несмещ.

$$D\tilde{\Theta}_2' = \frac{\cancel{4}(n+1)^2}{\cancel{4}(2n+1)^2} D[x_{max}] =$$

$$M x_{max}^2 = \int_{\Theta}^{2\Theta} \frac{n\cdot x^2\cdot(x-\Theta)^{n-1}}{\Theta^n}\,dx = \frac{4n^2+8n+2}{(n+1)(n+2)}\Theta^2$$

$$= \left(\frac{n+1}{2n+1}\right)^2\left(\frac{4n^2+8n+2}{(n+1)(n+2)} - \left(\frac{2n+1}{n+1}\right)^2\right)\Theta^2 =$$

$$= \left(\frac{n+1}{2n+1}\right)^2 \cdot \frac{n}{(n+1)^2(n+2)}\,\Theta^2 = \frac{n}{(2n+1)^2(n+2)}\cdot\Theta^2 \xrightarrow[n\to\infty]{} 0 \Rightarrow$$

$\Rightarrow$ состоят.

c) Эффективность

$$D\tilde{\Theta}_1 = \frac{\Theta^2}{27n} \qquad D\tilde{\Theta}_2' = \frac{\Theta^2 n}{(2n+1)^2(n+2)}$$

$$\frac{\Theta^2}{27n} < \frac{n\cdot\Theta^2}{(2n+1)^2(n+2)}$$

$$4n^3 + 12n^2 + 9n + 2 \le 0$$

$\Rightarrow$ Если $n \ge 3$ оценка $\tilde{\Theta}_2'$ эффективнее

d) Точный доверительный интервал

$\vec{x}_n$ выборка, $\xi \sim R[\Theta, 2\Theta]$

$$f(\vec{x}_n, \Theta) = \frac{x_{max}}{\Theta} - 1$$

$$P(f < t) = P(x_{max} < \Theta t + \Theta) = \left(F(\Theta t + \Theta)\right)^n =$$

$$= \left\{ F(x) = \begin{cases} 0, & x \le \Theta \\ \frac{x}{\Theta} - 1, & \Theta \le x \le 2\Theta \\ 1, & x > 2\Theta \end{cases} \right\}$$

$\alpha + \beta = 1$

$$t_1 = q_{\frac{\alpha}{2}} = \sqrt[n]{\frac{\alpha}{2}} = \sqrt[n]{\frac{1+\beta}{2}}$$

$$t_2 = q_{1-\frac{\alpha}{2}} = \sqrt[n]{1 - \frac{\alpha}{2}} = \sqrt[n]{\frac{1+\beta}{2}}$$

$$P\left(t_1 < \frac{x_{max}}{\Theta} - 1 < t_2\right) = \beta$$

$$\boxed{\frac{x_{max}}{1 + \sqrt[n]{\frac{1+\beta}{2}}} < \Theta < \frac{x_{max}}{1 + \sqrt[n]{\frac{1-\beta}{2}}}}$$

c) Асимптотический доверительный интервал

ОММ: $\tilde{\theta}_1 = \frac{2}{3}\bar{x} = \frac{2}{3}\tilde{\alpha}_1 = g(\tilde{\alpha}_1)$

$g(\alpha_1) = \frac{2}{3}\alpha_1 = \theta \qquad \nabla g = \frac{2}{3}$

$K_{11} = \alpha_2 - \alpha_1^2$

$\tilde{\mu}_2 = \tilde{\alpha}_2 - \tilde{\alpha}_1^2 = \frac{S^2(n-1)}{n}$

$$\frac{\sqrt{n}\,(\tilde{\theta}_1-\theta)}{\frac{2}{3}S\sqrt{\frac{n-1}{n}}} = \frac{3n(\tilde{\theta}_1-\theta)}{2S\sqrt{n-1}} \to N(0,1)$$

$$P\left(t_1 < \frac{3n(\tilde{\theta}_1-\theta)}{2S\sqrt{n-1}} < t_2\right) = \beta$$

$$\frac{2St_1\sqrt{n-1}}{3n} < \tilde{\theta}_1 - \theta < \frac{2St_2\sqrt{n-1}}{3n}$$

$$P\left(\tilde{\theta}_1 - \frac{2St_2\sqrt{n-1}}{3n} < \theta < \tilde{\theta}_1 - \frac{2St_1\sqrt{n-1}}{3n}\right) = \beta$$

№6

$$p(x) = \begin{cases} \frac{\theta-1}{x^{\theta}}, & x \geq 1 \\ 0, & x < 1 \end{cases}, \quad \theta > 1$$

а) ОМП:

$$L(\theta) = \prod_{i=1}^{n} p(x_i,\theta) = \prod_{i=1}^{n} \frac{\theta-1}{x_i^{\theta}} = (\theta-1)^n \cdot \prod_{i=1}^{n} x_i^{-\theta}$$

$$\ln L(\theta) = n\ln(\theta-1) - \theta\cdot\sum_{i=1}^{n}\ln x_i$$

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta-1} - \sum_{i=1}^{n}\ln x_i = 0 \to \theta = 1 + \frac{n}{\sum_{i=1}^{n}\ln x_i}$$

$$\frac{\partial^2 \ln L}{\partial \theta^2} = -\frac{n}{(\theta-1)^2} < 0 \Rightarrow \text{лок. макс} \Rightarrow \tilde{\theta} = 1 + \frac{n}{\sum_{i=1}^{n}\ln x_i} =$$

$$= 1 + \frac{1}{\overline{\ln x_i}}$$

б) Медиана: $\int\limits_1^{\hat{x}} p(x,\theta)\,dx = \frac{1}{2}$

$$\int\limits_1^{\hat x}(\theta-1)\,x^{-\theta}\,dx = (\theta-1)\left(\frac{x^{-\theta+1}}{-\theta+1}\right)\Bigg|_1^{\hat x} = 1-\hat x^{-\theta+1} = \frac12 \Rightarrow$$

$$\nabla g(\tilde\theta) = 2^{\frac{1}{\theta-1}}\cdot \ln 2\cdot(-1)\cdot\frac{1}{(\tilde\theta-1)^2} \qquad \Rightarrow \hat x = 2^{\frac{1}{\theta-1}} = g(\tilde\theta)$$

Проверим на непр. $I(\theta)=M\left[\left(\frac{\partial \ln p}{\partial\theta}\right)^2\right]$

$$\ln p = \ln(\theta-1)-\theta\cdot\ln x$$

$$\left(\frac{\partial \ln p}{\partial \theta}\right)^2 = \left(\frac{1}{\theta-1}-\ln x\right)^2$$

$$I(\theta)=\int\limits_1^{+\infty}\left(\frac{1}{(\theta-1)^2}-2\cdot\frac{\ln x}{\theta-1}+\ln^2 x\right)\frac{\theta-1}{x^\theta}\,dx=$$

$$=\int\limits_1^{+\infty}\left(\frac{x^{-\theta}}{\theta-1}-2\frac{\ln x}{x^\theta}+(\theta-1)\cdot\frac{\ln^2x}{x^\theta}\right)dx=$$

$$=\frac{1}{(\theta-1)^2}-2\cdot\left(\ln\frac{x^{1-\theta}}{1-\theta}\Bigg|_1^{+\infty}-\int\limits_1^{+\infty}\frac{x^{-\theta}}{1-\theta}\,dx\right)+$$

$$+\int\limits_1^{+\infty}\frac{(\theta-1)\ln^2x}{x^\theta}\,dx=\frac{1}{(\theta-1)^2}\to \text{непр. на } [1;+\infty)$$

$$\sqrt n\cdot\frac{(g(\hat\theta)-g(\theta))(\tilde\theta-1)}{2^{\frac{1}{\theta-1}}\cdot\ln 2}\rightsquigarrow N(0;1)$$

$$g(\tilde\theta)-\frac{1{,}96\cdot 2^{\frac{1}{\theta-1}}\cdot\ln 2}{\sqrt n\,(\tilde\theta-1)} < \hat x < g(\tilde\theta)+\frac{1{,}35\cdot 2^{\frac{1}{\theta-1}}}{\sqrt n\,(\tilde\theta-1)}$$

c)
$$\frac{\sqrt{n}\,(\hat{\theta}-\theta)}{\hat{\theta}-1} \rightsquigarrow N(0;1)$$

$$t_1 < \sqrt{n}\cdot \ln x \cdot \left(1+\frac{1}{\ln x}-\theta\right) < t_2$$

$$\Downarrow$$

$$1-\frac{t_2-\sqrt{n}}{\sqrt{n}\cdot \ln x} < \theta < 1-\frac{t_1-\sqrt{n}}{\sqrt{n}\,\ln x}$$